Multiple Linear Regression Modeling

Predicting Initial Hospital Days from Patient Variables

Research Question

Which variables cause a significant change in Initial Days spent in the hospital?

Analysis Goals

The goal of the data analysis will be to determine the set of independent variables which have the strongest relationship to the dependent variable, initial days. This could be used to predict how many days in the hospital will be needed based on factors such as what kind of services are needed, patient age, and many others. This data analysis will also provide a reduced linear regression model that can be used to estimate initial days, potentially allowing for deletion of irrelevant columns in long term data storage not used in any analysis, or even the Y variable, Initial Days.

Multiple Linear Regression Assumptions

The multiple linear regression model assumes that there is a linear relationship between the dependent variable and independent variables.

It also assumes that there is no multicollinearity, which would be multiple independent variables highly correlated with each other.

There is also the assumption of multivariate normality, meaning that there is a normal distribution of the residual values of the multiple linear regression model.

Finally, the model assumes homoscedasticity, or the idea that the residual values of the multiple linear regression model will be constant across the entire model.

Tool & Technique Justification

I will use Python for this data analysis because of the packages available, such as seaborn and matplotlib for visualization. Additionally, the ease of panda's capability to transform the data file into a dataframe and work with the data is extremely helpful.

Multiple linear regression is an appropriate technique because this will show us the relationship between a continuous numeric dependent variable, and multiple categorical and numeric independent variables. The dependent variable, initial days, is going to be analyzed from the perspective of how the independent variables in the dataset affect that value.

Data Cleaning

First, the data has many yes/no columns. Data analysis will be much easier if these are converted to 1/0. This is achieved by running a .replace function over the entire dataframe converting yes/no to 1/0.

We will also run a drop NA function followed by a fill NA function over the entire dataframe to ensure than any NA values are converted to the mean value for their column to allow for calculations.

Duplicates will be evaluated using .duplicate function, but no fields populate that are necessary to change.

Outlier analysis is also performed on all numeric columns, and outliers are replaced with the column mean.

All categorical columns are evaluated using .unique to ensure there are no duplicates, nulls, or values in need of conversion, and these all look good to proceed.

Summary Statistics

Summary statistics show us that the dependent variable, initial days, has a mean of 34.45, max of 71.98, and minimum of 1.0. Categorical columns such as initial admin list out each unique value for that column along with the count, showing emergency admission topping out at 5060. Complication risk is most likely to be medium at 4517 patients with this risk level. The average age for patients is 53.5 years, with a maximum of 89 years. Services is most often blood work, with 5265 occurrences. Patients are more likely to not have asthma than to have it, with 7107 patients reporting no. 7425 patients do not drink soft drinks compared to the 2575 who do. 7262 patients have diabetes. 7094 patients are overweight. Nearly half of patients have high blood pressure with 4090 such reports. Vitamin D support is most likely to not be present, with a large decrease in every level of increase.

```
Name: count, dtype: int64
Soft_drink
No     7425
Yes    2575
Name: count, dtype: int64
Diabetes
No     7262
Yes    2738
Name: count, dtype: int64
Overweight
Yes    7094
No     2906
Name: count, dtype: int64
HighBlood
No     5910
Yes    4090
Name: count, dtype: int64
vitD_supp
0    6702
1    2684
2     544
3      64
4       5
5       1
Name: count, dtype: int64
count    10000.000000
mean        34.455299
std         26.309341
min          1.001981
25%          7.896215
50%         35.836244
75%         61.161020
max         71.981490
Name: Initial_days, dtype: float64
```

```
Initial_admin
Emergency Admission      5060
Elective Admission       2504
Observation Admission    2436
Name: count, dtype: int64
Complication_risk
Medium    4517
High      3358
Low       2125
Name: count, dtype: int64
count    10000.000000
mean        53.511700
std         20.638538
min         18.000000
25%         36.000000
50%         53.000000
75%         71.000000
max         89.000000
Name: Age, dtype: float64
Services
Blood Work     5265
Intravenous    3130
CT Scan        1225
MRI             380
Name: count, dtype: int64
Stroke
No     8007
Yes    1993
Name: count, dtype: int64
Asthma
No     7107
Yes    2893
Name: count, dtype: int64
Soft_drink
```

Data Transformation

The categorical columns will need to be transformed into dummy columns and built into a new dataframe. This will be done using pandas' get dummies function and assigning these values to a dataframe that will be evaluated in feature selection.

Then we will use feature selection to reduce the initial model down to less than ten features.

Initial Model

```
In [3]:
   ...:
   ...: X0 = df[['Initial_admin', 'Income', 'TotalCharge', 'ReAdmis', 'Complication_risk', 'Age','Services', 'Stroke','Asthma', 'Diabetes',
'Overweight', 'HighBlood', 'vitD_supp']]
   ...: Y = df['Initial_days']
   ...: #build df with dummies
   ...: X = pd.get_dummies(data=X0,drop_first=True)
   ...: print(df.head())
   ...:
   ...: #one hot encode
   ...: X.replace(['Yes','No'],[1,0],inplace=True)
   ...: X.replace([True , False],[1,0],inplace=True)
   ...:
   ...: #Export cleaned data set
   ...: X.to_csv("C:/Users/velez/Downloads/InitialModelFeaturesProj1BJV.csv")
   ...:
   ...: #train on data
   ...: from sklearn.linear_model import LinearRegression
   ...: from sklearn import metrics
   ...:
   ...: #build LR
   ...: lr = LinearRegression()
   ...: lr.fit(X, Y)
   ...:
   ...: #compare Y values predicted to actual
   ...: y_pred = lr.predict(X)
   ...:
   ...: residuals= Y - y_pred
   ...: print("Y variable Residuals    :", residuals)
   ...: plt.scatter(residuals,y_pred)
   ...:
   ...: df2=pd.DataFrame(data={'Predictions':y_pred,'Actuals':Y})
```

Feature Selection

This will be done using forward feature selection, building the list starting up from nothing up to a point where the model can make accurate predictions. The criteria will be variables with a P-value below 0.05 will be selected. Using forward feature selection will add variables one at a time to nullify multicollinearity. Seven features are selected for the reduced model.

```
   ...: # printing the summary table
   ...: print(result.summary())
                                 OLS Regression Results
=======================================================================================
Dep. Variable:           Initial_days   R-squared (uncentered):                  0.986
Model:                            OLS   Adj. R-squared (uncentered):             0.985
Method:                 Least Squares   F-statistic:                         3.997e+04
Date:                Mon, 24 Feb 2025   Prob (F-statistic):                       0.00
Time:                        07:23:38   Log-Likelihood:                        -30708.
No. Observations:               10000   AIC:                                 6.145e+04
Df Residuals:                    9983   BIC:                                 6.157e+04
Df Model:                          17
Covariance Type:            nonrobust
=========================================================================================================
                                        coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------------------------
Income                            -4.791e-05   1.77e-06    -27.122      0.000   -5.14e-05   -4.45e-05
TotalCharge                           0.0094   3.66e-05    256.490      0.000       0.009       0.009
Age                                  -0.1120      0.002    -49.561      0.000      -0.116      -0.108
vitD_supp                            -0.8872      0.083    -10.731      0.000      -1.049      -0.725
Initial_admin_Emergency Admission    -8.8498      0.126    -70.213      0.000      -9.097      -8.603
Initial_admin_Observation Admission  -3.5818      0.144    -24.853      0.000      -3.864      -3.299
ReAdmis_Yes                           9.3957      0.184     51.020      0.000       9.035       9.757
Complication_risk_Low                 1.4443      0.141     10.238      0.000       1.168       1.721
Complication_risk_Medium              1.2094      0.114     10.572      0.000       0.985       1.434
Services_CT Scan                     -1.8338      0.165    -11.128      0.000      -2.157      -1.511
Services_Intravenous                 -1.6433      0.117    -14.087      0.000      -1.872      -1.415
Services_MRI                         -1.3826      0.277     -4.989      0.000      -1.926      -0.839
Stroke_Yes                           -1.0510      0.130     -8.068      0.000      -1.306      -0.796
Asthma_Yes                           -1.1526      0.115    -10.053      0.000      -1.377      -0.928
Diabetes_Yes                         -1.9643      0.117    -16.826      0.000      -2.193      -1.735
Overweight_Yes                       -3.1382      0.111    -28.400      0.000      -3.355      -2.922
HighBlood_Yes                        -2.5529      0.106    -24.157      0.000      -2.760      -2.346
```

```
Overweight_Yes                          -3.1382      0.111    -28.400      0.000      -3.355      -2.922
HighBlood_Yes                           -2.5529      0.106    -24.157      0.000      -2.760      -2.346
==============================================================================
Omnibus:                       33.406   Durbin-Watson:                   1.538
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               33.638
Skew:                           0.138   Prob(JB):                     4.96e-08
Kurtosis:                       2.936   Cond. No.                     2.66e+05
==============================================================================

Notes:
[1] R² is computed without centering (uncentered) since the model does not contain a constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[3] The condition number is large, 2.66e+05. This might indicate that there are
strong multicollinearity or other numerical problems.
```

```
    ...: print(f"Selected features: {selected}")
Selected features: ['TotalCharge', 'Initial_admin_Emergency Admission',
'Complication_risk_Medium', 'Complication_risk_Low', 'HighBlood_Yes', 'Diabetes_Yes',
'ReAdmis_Yes']
```

```
Name: Initial_days, Length: 10000, dtype: float64
   Predictions    Actuals
0    12.246223  10.585770
1    13.823244  15.129562
2     2.850439   4.772177
3     1.435756   1.714879
4     1.247289   1.254807
Reduced model R Square            : 0.9982407813284938
Reduced model Intercept           : -29.29550316374432
Reduced model Mean Absolute Error : 0.8949850712769856
Reduced model Mean Squared Error  : 1.217576743908555
Reduced Residual Standard Error   : 1.1034385999721756
Reduced model coefficients        : [ 0.01210144 -6.24495159  4.9814396   4.97155519 -1.37995826 -0.91328712
  0.36434658]
                                 OLS Regression Results
=======================================================================================
Dep. Variable:           Initial_days   R-squared (uncentered):                   0.986
Model:                            OLS   Adj. R-squared (uncentered):              0.985
Method:                 Least Squares   F-statistic:                          3.997e+04
Date:                Mon, 24 Feb 2025   Prob (F-statistic):                        0.00
Time:                        07:27:11   Log-Likelihood:                         -30708.
No. Observations:               10000   AIC:                                  6.145e+04
Df Residuals:                    9983   BIC:                                  6.157e+04
Df Model:                          17
Covariance Type:            nonrobust
=========================================================================================================
                                          coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------------------------
Income                               -4.791e-05   1.77e-06    -27.122      0.000   -5.14e-05   -4.45e-05
TotalCharge                              0.0094   3.66e-05    256.490      0.000       0.009       0.009
Age                                     -0.1120      0.002    -49.561      0.000      -0.116      -0.108
vitD_supp                               -0.8872      0.083    -10.731      0.000      -1.049      -0.725
Initial_admin_Emergency Admission       -8.8498      0.126    -70.213      0.000      -9.097      -8.603
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ReAdmis_Yes                              9.3957      0.184     51.020      0.000       9.035       9.757
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Prob(Omnibus):                  0.000   Jarque-Bera (JB):               33.638
```

```
4         5     C544523  ...     4     3

[5 rows x 50 columns]
Y variable Residuals    : 0      -1.660454
1       1.306318
2       1.921738
3       0.279123
4       0.007518
          ...
9995    0.593906
9996    1.225579
9997    0.307928
9998    1.042290
9999   -0.640585
Name: Initial_days, Length: 10000, dtype: float64
   Predictions    Actuals
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Method:                Least Squares   F-statistic:                          3.997e+04
Date:               Tue, 04 Mar 2025   Prob (F-statistic):                        0.00
Date:               Tue, 04 Mar 2025   Prob (F-statistic):                        0.00
Time:                       12:25:16   Log-Likelihood:                          -30708.
No. Observations:              10000   AIC:                                  6.145e+04
Df Residuals:                   9983   BIC:                                  6.157e+04
Df Model:                         17
Covariance Type:           nonrobust
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                                          coef    std err          t      P>|t|      [0.025      0.975]
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Prob(Omnibus):                  0.000   Jarque-Bera (JB):               33.638
Skew:                           0.138   Prob(JB):                     4.96e-08
Kurtosis:                       2.936   Cond. No.                     2.66e+05
==============================================================================

Notes:
```
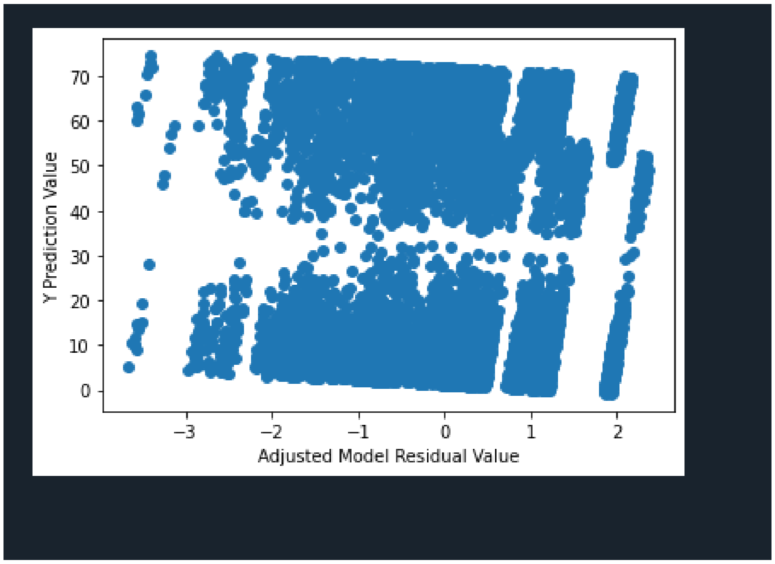

Model Comparison

The initial model and reduced model both have very similar R Square scores, Mean Absolute Errors, and Residual Standard Errors. This gives confidence that the reduced model is going to be nearly just as accurate as the initial model.

Residual Plot



This residual plot appears to have randomly scattered residual values. That would indicate this model is a strong and relevant fit to use on this data.

Residual Standard Error

Reduced model RSE = 1.103

Residual standard error is a metric that defines how well model predictions fit relative to actual values. The value of this model being 1.103 means that on average, actual values are 1.103 initial days off from prediction. This is very close and means the model is providing meaningful predictions.

```
Reduced model R Square             : 0.9982407813284938
Reduced model Intercept            : -29.29550316374432
Reduced model Mean Absolute Error  : 0.8949850712769856
Reduced model Mean Squared Error   : 1.217576743908555
Reduced Residual Standard Error    : 1.1034385999721756
Reduced model coefficients         : [ 0.01210144 -6.24495159  4.9814396   4.97155519 -1.37995826 -0.91328712
  0.36434658]
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Model:                            OLS   Adj. R-squared (uncentered):              0.985
Method:                 Least Squares   F-statistic:                          3.997e+04
Date:                Tue, 04 Mar 2025   Prob (F-statistic):                        0.00
Time:                        12:25:16   Log-Likelihood:                         -30708.
No. Observations:               10000   AIC:                                  6.145e+04
Df Residuals:                    9983   BIC:                                  6.157e+04
Df Model:                          17
Covariance Type:            nonrobust
```

Regression Equation

Y (Initial Days) = 0.0121(TotalCharge) , 6.24(Initial_admin_Emergency Admission) + 4.98(Complication_risk_Medium) + 4.97(Complication_risk_Low) , 1.37(HighBlood_Yes) ,0.913(Diabetes_Yes) + 0.36(ReAdmis_Yes)

Coefficient Interpretation

The coefficients for categorical columns in this model are simply going to be added into the equation if present, as they are multiplied by one if true. When any independent variable is false, it is simply zeroed out as it multiplies by the zero value indicating false from the dataframe.

For the continuous numeric variable in the reduced model, that coefficient of 0.0121 will be multiplied by the dollar value of total charge and added in.

The coefficients demonstrate that being brought in as an emergency admission can be expected to, on average, lessen the stay for a patient by 6.24 days compared to other admission reasons, all else equal. Having a medium complication risk adds 4.98 expected average days, while low complication risk adds 4.97. Having high blood pressure decreases the average expected stay by 1.37 days. Patients with diabetes can be expected to spend an average of 0.913 less days in their stay. Being readmitted averages to an increase of 0.36 days.

The higher absolute value coefficients for categorical columns will have a more significant impact on the outcome of the model, so their relative importance is clear.

Statistical & Practical Significance

The R square of this model is .9982, indicating this is a statistically relevant model to predict the Y value, initial days. This gives us the knowledge that the model estimates will be 99.82% accurate.

Practically, this model could be used to predict the amount of time patients will need to spend in the hospital, allowing for more accurate preparation of supplies and staffing for patients based on the most impactful model features. This could also be used as an alternative to storing large amounts of data, instead of storing the initial days column, that data could be erased and calculated as needed for a significant cost decrease. Some companies may also be able to determine that the columns not used in any relevant models are not worth the cost of storing long term.

Limitations

When analyzing this data functionally, hospitals will not have all of the independent variables quickly enough to accurately predict how many days will be necessary. This will be more helpful as general rules of thumb, such as the knowledge that a certain service would add 2.5 days to a patient's stay on average.

All statistical models are built using past data, this model did not predict or create that data, it is created using the outcome. This means this is retrofitted, applying and fitting perfectly to past data, but we have no understanding of whether future data will follow any of the same trends. There are always unseen factors that the model cannot account for, such as natural disasters, spread of disease, a change in availability of some alternative form of care.

Additionally, the data itself is likely not perfect. Patients may not have a good understanding of what they are being asked, people may answer differently based on mood, recency bias, beliefs, etc. There are many potential ways datapoints could be inaccurate when analyzing a large population.

Course of Action

Using the reduced model, it may be prudent to reduce the amount of data stored by the hospital over long periods of time. Data storage could be reduced to the columns needed for this and any other relevant models, saving on costs and easing processing. The hospital can now prepare for patients in real time by having an understanding of how these factors change the initial days value, as the hospital gains information on the patient during their stay, they can predict more and more what will be necessary.